

Part A · Multiple Choice

2 points each

1. **a) The color of the parent rock**

p. 152 — the five factors are parent rock composition, temperature, the amount and type of pressure, the fluids present (mostly water), and time.

2. **c) directed pressure or shear stress**

pp. 153–154 — because of plate tectonics, crustal pressure is usually greater in one direction, so new minerals grow with their long axes perpendicular to the squeezing. That directional fabric is foliation.

3. **d) Slate, phyllite, schist, gneiss**

p. 154 — slate breaks into flat sheets with microscopic crystals; phyllite is hotter and shows a micaceous sheen; schist has visible mica crystals; gneiss separates into colored mineral bands and usually has little or no mica.

4. **a) limestone**

p. 156 — the calcite crystals grow larger, destroying any original sedimentary textures and fossils. Quartzite, by contrast, is metamorphosed sandstone.

5. **b) speeds reactions up by helping ions move, and carries dissolved elements from place to place**

p. 153 — water does not necessarily change the outcome of a metamorphic process, but it lets reactions finish that might otherwise run out of time. It is also central to hydrothermal processes and mineral deposits.

6. **b) 30°C per kilometer**

p. 162 — in volcanic areas it steepens to 40–50°C/km, while along subduction zones the cold descending crust keeps it below 10°C/km.

Part B · True or False

1 point each

7. **TRUE**

p. 154 — foliation requires minerals that can grow or rotate into a preferred orientation, which is why mudrocks (rich in clay and mica) foliate so readily.

8. **FALSE**

The zone around an intrusion is very small — typically meters to tens of meters. Tens of thousands of square kilometers is the scale of regional metamorphism. (p. 161)

Part C · Short Answer

3 points each

9. **Contact metamorphism is local heating around an intruding magma body at shallow depth, without directed pressure, so the rock is not foliated. Regional metamorphism occurs over vast areas at depth, especially beneath mountain ranges at convergent boundaries, under strong directed pressure, so the rock is foliated.**

Rubric: 1.5 pts contact — heat from a magma body in the upper crust, a zone only meters to tens of meters wide, little or no directed pressure, producing non-foliated rocks such as hornfels, marble, and quartzite. 1.5 pts regional — deep burial over tens of thousands of square kilometers, greatest in the roots of mountain ranges such as the Himalayas at continent–continent convergent boundaries, with strong directional pressure producing foliated rocks. (pp. 160–161)

10. Most blueschist keeps subducting, turns into eclogite at about 35 km depth, and sinks into the mantle. It reaches the surface only where subduction was interrupted by some tectonic process — for example the Franciscan Complex around San Francisco.

Rubric: 1 pt — blueschist forms in subduction zones under very high pressure but relatively low temperature, since the cold descending oceanic crust does not heat up quickly (its blue color comes from the amphibole glaucophane). 1 pt — it normally continues down, converting to eclogite around 35 km and sinking into the mantle, never to return. 1 pt — it surfaces only where the subduction process was interrupted; accept the Franciscan Complex near San Francisco. (pp. 161–162)